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XDDT PART 1 3/15/73 JBL (XDDTA,16)

ASC=720051
CAC=720053
CBS=720056
CKS=720033
DSC=720050
EEM=724074
ESM=720055
IOH=730000
ISB=720052
LAT=762200
LEM=720074
LSM=720054
PPA=720005
PPB=720006
RPA=720001
RPB=720002
RRB=720030
SFT=660000
SZS=640000
TYI=720004
TYO=720003

/IN-OUT HALT

DBA=722061
DCL=720063
DIA=720061
DRA=722062
DWC=720062
ERM=720065
LRM=720064
RCC=723022
RCH=722022
RCK=720032
RNM=720066
RRC=722122
RRI=720037
RRO=720017
RSC=721122
RSM=720067
RTB=720035
SSB=724122
TCB=724022
TCC=725022

HALT=760400
TYIHNG=IOT 100
NETM1U=IOT 643
NETM1D=IOT 143
SUPPC=40 /PLACE TO PLANT INSTRUCTION
PC=35
DDTCOR=100000
L=0 R=1
DDM=(DDTCOR)
EIC=(DDTCOR+SGP)
EVC=(DDTCOR+EST)

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C20=(20)
C77=(77)
CI=(I)
CJ=(JMP)
C4=(4000000)

DEFINE DISP LC,UC
Z=LC-LSE 44
Z"T"1000 UC-LSE 44
TERMINATE DISP

DEFINE LETTER A,B
DISP A+LSE-44,B
TERMINATE LETTER

DEFINE SGOZE A,B,C
A"T"1600.+B"T"40.+C
TERMINATE SGOZE

DDTCOR/ JMP RLSE
JMP USERR

STAT, 0
DWANT, 0
DDTO, 0
DDTI, 0
DDTRPA, 0
DDTCLR, 0
DDTRRD, DDTCOR LSE
IOPDNE, 0
12, 0
DCONT, 0
14, 0
WTOP, 0
WHERE, 0
NETHND, 0
20, CLA

/CORE 17 LINK INTO CORE 10
/UNUSED

/UNUSED

/HALT PROCEDURE

LSM
SAS I DCONT
JMP ER5
JSP I IOPDNE
NETM1D

/IOP BUSY

26, HLT /MAY PUSH CONTINUE
27, JMP SYSUP /ENTRY FROM RESTART PROCEDURE

30, LIO (700000 /UNHANG THE DATA CHANNEL
RRO 500

HLT
JMP T+1
SYSUP, LAC I NETHND
SZA
NETM1U
JMP T+1

/NET BACK UP IF A NET USER EXISTS

LOWLIM=40
DDTCOR+4200/

LOW,	0	-0	/INITIAL SYMBOL TABLE
	0	DDTCORE	
	SQOZE	0, 27, 46	MSK
	SQOZE	0, 36, 46	EST
	SQOZE	0, 15, 46	MEM
	0		400000

/OPERATE GROUP (IN EVENT TIME ORDER)

	SQOZE	15, 15, 23	CCI	
	SQOZE	26, 13, 32	LAP	
	SQOZE	26, 13, 36	LAT	
	SQOZE	15, 26, 15	CLC	
	SQOZE	15, 26, 13	CLA	
	SQOZE	15, 26, 23	CLI	
	SQOZE	26, 13, 32	OPR	100 /LAT=CLA OPR
	SQOZE	26, 13, 36	OPR	2000 /LAT=CLA OPR
	SQOZE	15, 27, 13	CMA	
	SQOZE	15, 27, 23	CMI	
	SQOZE	22, 26, 36	HLT	
	SQOZE	35, 41, 32	SWP	
	SQOZE	26, 13, 23	LAI	
	SQOZE	26, 23, 13	LIA	
	SQOZE	35, 36, 20	STF	
	SQOZE	15, 26, 20	CLF	
NOPCOD,	SQOZE	30, 31, 32	NOP	
	SQOZE	31, 32, 34	OPR	

/SPECIAL OPERATE GROUP (IN EVENT TIME ORDER)

SQOZE	26, 20, 23	LFI
SQOZE	26, 23, 20	LIF
SQOZE	35, 44, 26	SZL
SQOZE	35, 15, 23	SCI
SQOZE	35, 15, 20	SCF
SQOZE	15, 26, 26	CLL
SQOZE	23, 20, 23	IFI
SQOZE	23, 23, 20	IIF
SQOZE	35, 15, 27	SCM
SQOZE	15, 27, 26	CML
SQOZE	23, 16, 13	IDA
SQOZE	23, 16, 15	IDC
SQOZE	13, 13, 23	AAI
SQOZE	23, 13, 23	IAI
SQOZE	42, 13, 23	XAI
SQOZE	35, 32, 31	SPO

/SKIP GROUP

SQOZE	15,	26,	31	CLO
SQOZE	35,	32,	33	SPQ
SQOZE	35,	44,	27	SZM
SQOZE	35,	30,	23	SNI
SQOZE	35,	32,	23	SPI
SQOZE	35,	44,	31	SZO
SQOZE	35,	27,	13	SMA
SQOZE	35,	32,	13	SPA
SQOZE	35,	44,	13	SZA
SQOZE	35,	44,	20	SZF
SQOZE	35,	44,	35	SZS
SQOZE	35,	25,	32	SKP

/STANDARD INSTRUCTIONS

SQOZE	0,	0,	23	I
SQOZE	13,	16,	16	ADD
SQOZE	13,	30,	16	AND
SQOZE	13,	35,	15	ASC
SQOZE	15,	13,	15	CAC
SQOZE	15,	13,	26	CAL
SQOZE	15,	14,	35	CBS
SQOZE	15,	25,	35	CKS
SQOZE	16,	13,	15	DAC
SQOZE	16,	13,	32	DAP
SQOZE	16,	14,	13	DBA
SQOZE	16,	15,	22	DCH
SQOZE	16,	15,	26	DCL
SQOZE	16,	23,	13	DIA
SQOZE	16,	23,	31	DIO
SQOZE	16,	23,	32	DIP
SQOZE	16,	23,	40	DIV
SQOZE	16,	34,	13	DRA
SQOZE	16,	35,	15	DSC
SQOZE	16,	41,	15	DWC
SQOZE	16,	44,	27	DZM
SQOZE	17,	17,	27	EEM
SQOZE	17,	34,	27	ERM
SQOZE	17,	35,	27	ESM
SQOZE	23,	16,	42	IDX
SQOZE	23,	31,	22	IOH
SQOZE	23,	31,	34	IOR
SQOZE	23,	31,	36	IOT
SQOZE	23,	35,	14	ISB
SQOZE	23,	35,	32	ISP
SQOZE	24,	16,	13	JDA
SQOZE	24,	27,	32	JMP
SQOZE	24,	35,	32	JSP
SQOZE	26,	13,	15	LAC
SQOZE	26,	13,	41	LAW
SQOZE	26,	15,	22	LCH
SQOZE	26,	17,	27	LEM
SQOZE	26,	23,	31	LIO
SQOZE	26,	34,	27	LRM

SQOZE 26, 35, 27
 SQOZE 27, 37, 26
 SQOZE 32, 32, 13
 SQOZE 32, 32, 14
 SQOZE 34, 13, 26
 SQOZE 34, 13, 34
 SQOZE 34, 15, 15
 SQOZE 34, 15, 22
 SQOZE 34, 15, 25
 SQOZE 34, 15, 26
 SQOZE 34, 15, 34
 SQOZE 34, 23, 26
 SQOZE 34, 23, 34
 SQOZE 34, 30, 27
 SQOZE 34, 32, 13
 SQOZE 34, 32, 14
 SQOZE 34, 34, 14
 SQOZE 34, 34, 15
 SQOZE 34, 34, 23
 SQOZE 34, 34, 31
 SQOZE 34, 35, 15
 SQOZE 34, 35, 27
 SQOZE 34, 36, 14
 SQOZE 35, 13, 16
 SQOZE 35, 13, 26
 SQOZE 35, 13, 34
 SQOZE 35, 13, 35
 SQOZE 35, 15, 26
 SQOZE 35, 15, 34
 SQOZE 35, 20, 36
 SQOZE 35, 23, 26
 SQOZE 35, 23, 34
 SQOZE 35, 35, 14
 SQOZE 35, 37, 14
 SQOZE 36, 13, 16
 SQOZE 36, 15, 14
 SQOZE 36, 15, 15
 SQOZE 36, 43, 23
 SQOZE 36, 43, 31
 SQOZE 42, 15, 36
 SQOZE 42, 31, 34
 SQOZE 0, 42, 42
 SQOZE 0, 2, 35
 SQOZE 0, 3, 35
 SQOZE 0, 4, 35
 SQOZE 0, 5, 35
 SQOZE 0, 6, 35
 SQOZE 0, 7, 35
 SQOZE 0, 10, 35
 SQOZE 0, 11, 35
 SQOZE 0, 12, 35

SGP,

EST,

DDTCORE LOW

LSM
 MUL
 PPA
 PPB
 RAL
 RAR
 RCC
 RCH
 RCK
 RCL
 RCR
 RIL
 RIR
 RNM
 RPA
 RPB
 RRB
 RRC
 RRI
 RRO
 RSC
 RSM
 RTB
 SAD
 SAL
 SAR
 SAS
 SCL
 SCR
 SFT
 SIL
 SIR
 SSB
 SUB
 TAD
 TCB
 TCC
 TYI
 TYO
 XCT
 XOR
 XX
 1S
 2S
 3S
 4S
 5S
 6S
 7S
 8S
 9S

```

RLSE,      JSP I DDTRRD
LSE,        JSP LCC           /LISTEN LOOP
LSE+1,      CLC"U"CLF 7
            DAC SP2
            DAC OPN
LSS,        CLC
            DAC CHI
            DAC SP1
LSP,        DZM WRD
SSN-1,      LAC CUN           /(IOR)
SSN,        DIP SGN
BSLASH,     DZM DNM
            DZM SYL
N2,         DZM SYM L         /LOOP FOR NEXT SYLLABLE
            DZM SYM R
            DZM FSM
            CLC"U"CLF 4
            DAC LET
            DAC CC
LSR,        LIO SK1           /LOOP PER CHAR.  CHEAP PLACE...
            DIO WEA           /...TO DO INITIALIZATION,
            LAW VF5+1
            DAP VF5
            LAW LWT
            DAP BAX
            JSP I DDTI        / TYPE-IN, HANG IF NO CHARS
            RAL 6S
            DAC CH
CDB,        LAW DTB           /USED AS DTB
            DAC FL1           /NON-ZERO
CAD,        ADD CH           /USED AS ADD
            DAP .+1
LSL,        LIO .
            ISP CAS
            JMP UPPER
LSU,        SZF I 4           /"QUOTE" MODE?
            JMP NOTQM        /NO
            IDX CC
            SAS (4
            JMP LN4
QUOTE2,     LAC FSM
            JMP N1

```

/NO-EVAL ROUTINES

ULC,	LAW I 2 DAC CAS JMP LSR	/77 SEEN
EOT,	CLC DAC MSK DAP UL DZM LL LAC (DCH DAC MEM DAC TAS LAW 8. DAC RADIX DZM USER DZM NUM LAW PI DAP PNS LAW PEV DAP PA1 JMP LSE+1	/EOT SEEN. RESET EVERYTHING FOR LARRY
CHH,	LIO LET LAC CHI SPI"U"SMA JMP SHALL LAW CHTBL	/NUMBER-SIGN LOOKUP
SEEK, /AND YE, /SHALL /FIND ERR,	DAP .+1 LAC . SZA JMP CHH1 LAW CHARAC R? JDA TYS JMP LSR	/PRINT Q-MARK. IGNORE LAST CHAR.
XCL,	LAW XCTBL JMP SEEK	/EXCLAMATION MARK
DOT,	LIO CC LAC LOC SPI I LAC DNM DAC SYL DAC DNM LAW 44 DAC T2 JMP LN1	/PERIOD
QUO,	LAC FSM SZA JMP N1 STF 4 JMP LSR	/" MEANS TAKE AS INTERNAL

Q,	LAC LWT JMP N1	/"Q" IS LAST QUANTITY TYPED
DAQ,	LAW 7777 AND LWT JMP COM+1	/<DEFINES SYM AS ADDRESS OF Q
COM, COM+1,	LAC LOC DAC DF1 DZM FL1	/COMMA DEFINES SYM AS LOC
DEF, SK1,	LAC LET SZA JMP ERR JSP DE JMP PN2	/DEFINE SYMBOL
DEL,	LAW 3 JDA TYS	/#,DELETE
DEL1,	JMP PN2	/ END OF NO EVAL ROUTINES
/SET OUTPUT RADIX		
RADX,	SNI"U"SZM SAD (1 JMP ERR DAC RADIX JMP LSE	/SET RADIX
/ARITHMETIC FUNCTIONS		
PLS,	LAC CAD JMP SSN	/PLUS SIGN
MIN,	SPI DIO WRD LAC CSU JMP SSN	/MINUS SIGN /(SUB)
ISC,	LAC CAN JMP SSN	/(AND) LOGICAL "AND".
UNI,	JMP SSN-1	/LOGICAL "OR"

/PRINT "Q" IN FORM SPECIFIED

```

EQL,      DAC LWT      /PRINT INTEGER IN CURRENT RADIX
          JSP LCT
          JDA OPT
PN2,      JSP LCT
          JMP LSS

```

```

DEC,      DAC LWT      /PRINT AS DECIMAL INTEGER
          LAC RADIX
          DAC T
          LAW 10,
          DAC RADIX
          JSP LCT
          JDA OPT
          LAC T
          DAC RADIX
          JMP PN2

```

```

ARW,      DAC LWT      /PRINT AS INSTRUCTION
          JSP LCT
          DAC PI      /JDA PI, BUT DON'T SET FLAG 2
          LAW PN2
          STF 3        /FORCE SYMBOLIC PRINTOUT
          JMP PI+2

```

```

PBX,      DAC LWT      /PRINT AS INTERNAL
          JSP LCT
          JDA TYSA
          JMP PN2

```

```

SQP,      DAC LWT      /PRINT AS SQOZE CODE
          JSP LCT
          DAC SYM R
          DZM SYM L
          LAW PN2
          DAP SPX
          JMP SPTLIM

```

/SET CURRENT CORE

```

CORE,     SNI I
          JMP ERR
          RAR 6S
          AND (170000
          DIP MEM
          DIP TAS      /YES,VIRGINIA, IT'S NECESSARY
          JMP LSE

```

/REGISTER EXAMINATION AND CHANGE

```

VBR,      AND (177777      / ;
          DIP TAS
          SPI
          JMP VBR1
          DIP MEM
          JMP TA5

VBR1,     AND (170000
          SZA
          BRING1           /CALL BRING WITHOUT CHANGING MEM
          JMP VBR1X

BAC,      LAW OPT           /OPEN BRACKET (BAR=CONSTANT)
          JMP BAS+1

BAI,      LAW TYSA         /CONTROL "I" (BAR-INTERNAL)
          JMP BAS+1

BAS,      LAW PI           /CLOSED BRACKET (BAR-SYMBOLIC)
BAS+1,    DAP BAX
VBR1X,    LAC OPT
BAR,      SPI
          JMP TA6
          LIO SP1
          DIO SP2
          SPI
          JMP TA5
          AND (177777)
TR3,      DAC SP2
SP5,      JSP LCT
          LAC I SP2
          JMP TA7

CR,       DAC LWT           /LINE FEED
          JDA MRF
          LAW 7715
          JDA TYS
          JMP LSE+1

UC8,      JDA MRF           /> MEANS MAKE CORR. AND OPEN REGISTER
          JMP TA6

```

BS,	JDA MRF	/BACKSPACE
	LIO SP2	
	SPI I	
	JMP BS2	
	IDX LOC	
BS+5,	DAC LWT	
	JMP TA3+2	
FS,	JDA MRF	/ARROW UP (FORWARD SPACE)
	JSP LCC	
	LAW I 1	
	LIO SP2	
	SPI I	
	JMP FS1	
	ADD LOC	
	JMP BS+5	
TAB,	JDA MRF	/MEMORY FIELD SWITCH
TA3,	DAC LWT	
	JSP LCC	
TA3+2,	JDA PAD	
	LAW CHARAC R/	
	JDA TYS	
TA5,	DZM LOC	
	DAP LOC	
	LIO MEM	
	DIO TAS	
TA6,	LIO C4	
	DIO SP2	
	DAP TAS	
	JSP LCT	
	BRING	
	LAC I TAS	
TA7,	DAC LWT	
BAX,	JDA .	/PI, OPT OR LWT
	DZM OPN	
	JMP PN2	

/SYMBOL ROUTINES

VAL,	DAC DF1 JMP LSS	/OPEN PAREN, SETS UP VALUE FOR DEFINITION
TBL,	JSP I DDTCLR JMP RDNY DZM FL1 JSP SOI	/READ SYMBOL PUNCH
MR1,	JSP GWD DAC SYM L JSP GWD AND (177777 DAC SYM R IOR SYM L SZA JMP TBL1	
TBN,	JSP LCT LAC EST JDA OPT	
TBM,	JSP RBK JMP TBM	/SKIPS REST OF TAPE
KIL,	SPI JMP KI3	/KILL SYMBOL(S)
KI2,	LIO LET SPI JMP ERR	
KI1,	LAC I EV2 IOR KI1 DIP I EV2 JMP LSE	/USED AS LAC I

/ZERO CORE

PUL,

SPI
JMP ERR
DAC FA
JMP LSS

/LOWER LIMIT SETUP

ZRO,

JSP OK
LAW 7777
SPI
DAC WRD
DIP WRD
AND FA
SPI
CLA
IOR MEM
DAC T
LIO NUM
BRING

/ZERO CORE TO CONTENTS OF M#+3

ZR2,

SUB MEM
SUB WRD
SZM
JMP LSE
DIO I T
IDX T
JMP ZR2

/SEARCHES

EAS,	LAW EA1 JMP WS	/EFFECTIVE ADDRESS SEARCH
NWS,	LAC SK2 DAC WEA	/NOT WORD SEARCH
WDS, WS,	LAW WS1 SPI JMP ERR DAP WS2 JSP LCC LAC USER DAC MWSU DZM MWSXU	/WORD SEARCH
MWSNU,	DZM MWSFTI SZA I SAS MEM JMP NOTMWS LIO MWSXU IDX USER SAD (100 JMP LOOK1 ADD STAT DAC EAS1 LAW I 7777 AND I EAS1 SAD (760000 JMP WS3+5	/MOBY WORD SEARCH: IF USER=MEM=0, SEARCH... /...ALL USERS FOR WRD. /RESTORE USER; EXIT TO LSE.
NOTMWS,	BRING LAC LL IOR MEM DAC T DIP T2	
WS4,	DZM SYM DAP T2 LAC I T2	
WS2,	JMP .	/EA1 OR WS1

/ROUTINES CONCERNED WITH TIME SHARING

```

LOOK,      SPI                /SET USER BEING EXAMINED
           JMP CREATE         /GO CREATE NEW USER
           LIA
           SUB (100)
           SMA
           JMP ERR
           LAC STAT
           AAI
           DAC T
           LAW I 7777
           AND I T
           SAD (760000)
           JMP USERR
LOOK1,     DIO USER
           JMP LSE

HOLD,      SPI I
           JMP HALT1         /HALT USER
           LAW (IOR (LAC))    /PREVENT USER FROM HALTING
           JMP HOLD1

FREE,      LAW (AND (-LAC))    /ALLOW USER TO HALT
HOLD1,     DAP HOLD2         /SET OR CLEAR HELD BY XDDT BIT
           JSP NOTU0
           SPI I
           JMP ERR          /WORKS FOR CURRENT USER ONLY
           BRING1
           LAC I C4         /(4000000)
HOLD2,     XCT .
           DAC I C4
           JMP LSE

```

```

SETPTR,  SPI T           /FREE TT PTRS
          SPA
          JMP ERR
          SAL 2S
          ADD (141000
          DAC T
          SUB (141000+40"T"4)      /40 TT'S
          SMA
          JMP ERR
          LAW 101
          LSM
          DAC I T
          IDX T
          LIO I T      /HAD BETTER BE CORRECT
          IDX T
          DIO I T
          IDX T
          LAC I T
          DZM I T
          ESM
          JDA OPT
          JMP LSE

```



```

BGN,      BRING1      /START USER
          SPI
          JMP BGN3      /NO ARG
          AND (170000
          SZA I
          JMP BGN1      /CORE 0
          SZS I 60
          JMP ER4
BGN1,     JSP NOTU0
          LIO OPT
BGN2,     DIO I (PC    /ENTRY FROM "STRUP"
BGN3,     LAC STAT
          ADD USER
          DAC T
          LAW 1
          DAP I T      /HIGH QUEUE
          JMP LSE

XEC,      SNI I 60     /EXECUTE ARG
          JMP ER4      /SS6 NOT UP
          DAC .+1
T,        0
          NOP
          JMP LSE

/READ OR VERIFY PAPER TAPE

SVFY,     LAW VF2      /SPECIAL VERIFY
          DAP VF5
VFY,      SPI I        /VERIFY CORE AGAINST BINARY TAPE
          JMP ERR
          JSP I DDTCLR
          JMP RDNY
          JSP LCC
          LAC RB2      /(SAD I)
          JMP RD1

```

REPEAT IIF VP .-LSE+44-1000, PRINT ,MYSTIC 1000 EXCEEDED.

```

RD,      SPI I          /READ BINARY TAPE INTO CORE
        JMP ERR
        JSP OK
        JSP I DDTCLR
        JMP RDNY
        LAC BS1         /(DAC I)
RD1,     DIP VF4
        JSP SOI
        BRING
VF1,     LAC MEM
        DIP T
        LAC T
        SUB LL
        SUB MEM
        SPA
        JMP VF2
        ADD LL
        SUB UL
        SZM
        JMP VF2
VF4,     LAC I LA
        T               /DAC I OR SAD I
        JMP VF2
        SAS (XX)
        SAD NUM         /SPECIAL VERIFY IGNORES PGM MODIFICATION
        JMP VF5
        LAW I 7777
        AND I T
        ADD T
        SUB MEM
        SAD I LA        /IGNORE "." IN ADR IF SPECIAL VFY
VF5,     JMP .          /JMP .+1 OR JMP VF2
        JSP PAC
        JSP LCT
        LAC I LA
        JDA LWT
        JSP LCC
VF2,     IDX T
        IDX LA
        SAD RB1
        JSP RBK
        JMP VF1

```

/BRANCHES. PLACED HERE TO SAVE ROOM IN DISPATCH AREA.

/REST OF LISTEN LOOP

```

NOTQM,    CLA
          RCL 9S
          DAC T2
          SUB (44
          SPA
          JMP LN
          ADD TLS      /(JMP LSE)
          DAP LSX
          SUB (JMP DEL1 /LAST NO-EVAL ROUTINE
          SPQ
          JMP LSX
          LAW SYL
          LIO LET
          SPI I        /SKIP IF LETTER NOT SEEN
          JSP EVL
          JMP EV4
          LAW CHARAC RU /IGNORE INPUT AND START OVER
ER2,      JDA TYS
          JMP LSS

EV4,      DAP SGN
          LAC WRD
SGN,      XX          /OPERATOR AND SYLLABLE ADDR.
          DAC WRD
          LIO CHI
          SPI
          LAC LWT
          DAC OPT
LSX,      JMP .        /I, O. MINUS IF NO ARG.  ARG IN AC.

UPPER,    JSP TYP OUT
          XCT LSL
          RIL 9S
          JMP LSU

LN,        ADD (44-12  /LETTER-NUMBER LOGIC
          SPA
          JMP N
          DZM LET
LN1,      DZM CHI
          IDX T2
          IDX CC
          SAS (4
          JMP LN3
          LIO SYM R
          DIO SYM L
          DZM SYM R

```

LN3, SUB (6
 SZM
 JMP LSR
 LAC SYM R
 RAL 2S
 ADD SYM R
 RAL 3S
 ADD T2
 DAC SYM R
 LAW T 4
 ADD CC

LN4, SMA
 JMP LSR
 LAC FSM
 RAL 6S
 ADD CH
 DAC FSM
 LAW 77
 SAD CH
 JMP ULC
 LAW 3
 SZF 4
 SAS CC
 JMP LSR
 JMP QUOTE2

/ENTRY FROM QUOTE

N, LAC SYL
 RAL 3S
 CUN, IOR T2
 DAC SYL
 LAC DNM
 RAL 2S
 ADD DNM
 RAL 1S
 ADD T2
 DAC DNM
 JMP LN1

/USED AS IOR

/REST OF NUMBER-SIGN LOOKUP

SHALL, LAC SYL
 SPA
 JMP ERR
 SAL 2S
 SUB (100"T"4)
 SMA
 JMP ERR
 ADD (100"T"4+141000
 JMP DDS

CHH1, SAD SYM R
 JMP FIND
 IDX YE
 IDX YE
 JMP YE

FIND, IDX YE
 DAP .+1
 JMP .

STATUS, LAC STAT
 ADD USER
 JMP DDS

C, LAW MEM
 JMP F+1

M, LAW MSK
 JMP F+1

F, LAW EST
 F+1, IOR DDM
 DDS, DAC SP1
 N1, DZM CHI
 DAC SYL
 DAC DNM
 JMP N2

/MODE CHANGING ROUTINES

LOT, LAW TYSA
 JMP CNS+1

SMB, LAW PI
 JMP CNS+1

CNS, LAW OPT /SYMBOLIC-CONSTANT-FLEXO SWITCH SETUP
 CNS+1, DAP PNS
 JMP LSE

OAD, LAW PVL
 JMP RAD+1

RAD, LAW PEV /OCTAL-RELATIVE SWITCH SETUP
 RAD+1, DAP PA1
 TLS, JMP LSE /USED AS LSE

/REST OF BS

BS2, IDX SP2
 SP3, DAC LWT
 DIP MEM
 JDA PAD
 LAW CHARAC R/
 JDA TYS
 LAC TAS
 DIP MEM
 JMP SP5

/REST OF FS

FS1, ADD SP2
 DAP SP2
 JMP SP3

/REST OF KIL

KI3, LAC (DDTCOR LOW
 DAC EST
 JMP LSE

/REST OF TBL

TBL1, JSP GWD
 DAC DF1
 LAC SYM L
 LIO SYM L
 RIL 1S
 SMA"U"SPI
 JMP MR1
 AND (177777
 DAC SYM L
 JSP DE
 JMP MR1

/REST OF EFFECTIVE ADDRESS SEARCH

EA1, DAC EAS1 /SAVE INSTRUCTION
 LIA
 AND (770000
 SAD (CAL
 JMP EA4
 SAD (JDA
 JMP EA6
 SAD (LAW I
 JMP WS3
 AND (760000
 SAS (OPR /FLUSH THESE
 SAD (SPO
 JMP WS3

	SAS (SKP	
	SAD (SFT	
	JMP WS3	
	SAD (IOT	
	JMP WS3	
	SAS (DCH	
	SAD (LCH	
	JMP EA2	
	LAI	
EA3.,	AND CI	/IS THERE AN I BIT
SK1.,	SZA	
	JMP EA2	/YES
	LAC EAS1	
	AND (760000)	
	SAD (XCT)	
	JMP EA11	
	SAS (LCH	
	SAD (DCH	
	JMP EA5	
WS8,	LAW 7777	/NO, AND OFF INSTRUCTION PART OF FIRST WORD
	AND I T2	
WS1,	XOR WRD	/COMPARE
CAN,	AND MSK	/USED AS AND
WEA,	XX	/SZA OR SZA I
	JMP WS3	/SETUP NEXT WORD
WS6,	LAC MWSU	
	SAD MEM	
	SAS MWSFTI	
	JMP NOTMW1	
	IDX MWSFTI	
	JSP LCC	
	LAC USER	
	DAC MWSXU	
	JDA OPT	
	LAC (FLEXO "L"	
	JDA TYS	
	JSP LCC	
NOTMW1,	LAW LCC	
PAC,	DAP PAX	/HIT, PRINT OUT
	LAC T	
	DAP LOC	
	JDA PAD	
	LAW CHARAC R/	
	JDA TYS	
	JSP LCT	
	LAC I T	
	JDA LWT	
PAX,	JSP .	

WS3,	IDX T	/INDEX
	SUB MEM	
	SUB UL	
	SPQ	
	JMP WSNW	
WS3+5,	LAC MWSU	
	SAS MEM	
	JMP LSE	
	JMP MWSNU	
WSNW,	LAC T	
	JMP WS4	
EA4,	LAC WRD	/CAL FINDS 100,101, AND ITS ADDRESS
	SAS (100	
	SAD (101	
	JMP WS6	
	JMP WS8	/CHECK ADDRESS
EA6,	JSP EAS1+1	/JDA,CHECK THING ADDRESSED
	IDX EAS1	/CHECK ADDRESS+1
EA7,	AND (7777	
	JMP WS1	
EA2,	LAW EA3.	/GET REG, REFERENCED FOR I,LCH,DCH
	JMP EA12	
EA11,	LAW EA1	
EA12,	DAP EA13	
	LAI	
	JDA EAS1	
	IDX SYM	
	SAD (10.	
	JMP WS3	
	LAI	
	DAP T2	
	LAC I T2	
	LIA	
EA13,	JMP .	
EA5,	LAI	/LCH,DCH PTR
	LIO EAS1	/INSTR.
	RIL 5S	
	SPI	
	IDC	
	JDA EAS1	/CHECK LOC. REFERENCED BY POINTER
	LAC T2	
	JMP EA7	/CHECK LOC OF POINTER
EAS1,	Ø	/SPECIAL COMPARE
EAS1+1,	DAP EAS2	
	LAW 7777	
	AND EAS1	
	XOR WRD	
	AND MSK	
	SZA	
EAS2,	JMP .	/MISS,RETURN

JMP WS6

/HIT, PRINT AND QUIT

/REST OF "LOOK"; CREATE NEW USER.

```

CREATE,      LIO (500
             LAC (140000
             LSM
             IOR I WTOP
             DAC T
             SUB WHERE
             DAC OPT
             SUB (100)
             SMA
             JMP ER5
             IDX I WTOP
             DIO I T
             ESM
             LAC OPT
             DAC USER
             JDA OPT
             BRING1
             LAW 11
             DAC T
             LIO NUM
STRTU1,      DIO I T
             IDX T
             SAS CI
             JMP STRTU1
             LAC (TYIHNG
STRTUP,      BRING1
             DAC I (SUPPC
             CLA
             DIP I C4      /CLEAR "BITS" IN CASE HALTING
             LIO (SUPPC
             JMP BGN2

```

/REST OF "HOLD"; HALT CURRENT USER.

```

HALT1,      SAS (1
             JMP ERR
             JSP NOTU0
             LAC (HALT
             JMP STRTUP

```

START